

Jordy Ruiz

Research/Software Engineer

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Profile

I am a French research/software engineer based in London since 2019, with Ph.D and industrial experience in low-level program analysis of embedded systems: I analyse programs both directly in binary form and in LLVM IR [1]. I bring a strong appreciation for mathematical rigour, well-structured programming, and clever use of abstractions. I also enjoy mentoring and working within cross-disciplinary teams.

Work

Software Team Lead at StatInf – 2 years

prev. **Research Engineer** at StatInf – 4 months

2022/10 –

- R&D for a hybrid static/statistical analysis of binary embedded systems programs
- Led a small software team (4–7 engineers), built from the ground up, mentoring collaborative development
- Management using Agile methods, interfacing with R&D, Product and Corporate teams
- In-house disassembly and program analysis plugins for our clients' architecture needs (TMS320C28, MPC5554 PPC, ARMv7+Thumb-2, ARMv9...)
- Led the development and go-to-market of StatInf's key product **RocqStat™**, including:
 - GUI design and implementation in Qt;
 - Backend development for our binary code and software trace analyses;
 - Securing our patented EVT algorithms;
 - Docker containerization;
 - CI/CD pipeline setup with integration testing;
 - Go-to-production processes and product demonstrations;
 - Iterative collaboration on product specifications.

[#C++](#) [#Python](#) [#QML](#) [#ARM-assembly](#) [#TI-assembly](#) [#PPC-assembly](#) [#embedded-systems](#)
[#statistical-analysis](#) [#static-analysis](#) [#software-traces](#) [#team-management](#)
[#Agile-methods](#) [#collaborative-development](#) [#CI-CD](#) [#Github-Actions](#) [#Docker](#) [#Deployment](#)

Research Associate at Imperial College London

2019/10 – 2022/09

Automating test generation using “chopped” symbolic execution in KLEE, through program analysis of LLVM IR code.

with Pr. Cristian Cadar

[#C++](#) [#LLVM](#) [#KLEE](#) [#patch-testing](#) [#static-analysis](#) [#dynamic-analysis](#) [#concolic-execution](#)

Postdoctoral fellowship at Université de Lille

2018/06 – 2019/06

Detection of arrays in assembly code for static timing analysis

with Pr. Giuseppe Lipari

[#C++](#) [#ARM-assembly](#) [#OTAWA](#) [#PPL](#) [#static-analysis](#) [#abstract-interpretation](#) [#WCET](#)

Internship at Université Toulouse III

2013/07

Co-inductive reasoning for the transformation of deterministic automata.

Complete formal proof of Brozowski's algorithm¹ in Coq.

with Dr. Ralph Matthes

[#Coq](#) [#Haskell](#) [#formal-proofs](#) [#category-theory](#)

¹Based on the categorical proof by Bonchi et al., *Brzowski's Algorithm (Co)Algebraically* (2012)

Education

Université Toulouse III

Ph.D. at Université Toulouse III

2014/10 – 2017/12

Thesis: Identifying data flow properties to improve worst-case execution time estimations

Analysed actual critical embedded applications (Continental, DEBIE-1, PapaBench...)

with Pr. Christine Rochange

[#C++](#) [#ARM-assembly](#) [#OTAWA](#) [#static-analysis](#) [#embedded-systems](#) [#infeasible-paths](#)
[#abstract-interpretation](#)

Masters in Computer Science	2012 – 2014
Second year: Critical Software & Distributed systems (ranked 2/8)	
First year: Artificial Intelligence & Pattern Recognition (ranked 1/15)	
Bachelor in Fundamental Mathematics	2009 – 2012
Bachelor in Computer Science	2009 – 2012
Parallel studies and simultaneous graduation from two bachelors.	

Publications in journals, international conferences, and workshops

Relational Abstract Interpretation of Arrays in Assembly Code.	2021/12
C. Ballabriga, J. Forget, J. Ruiz	
In: <i>Formal Methods in System Design</i> , Volume 59. (regular paper)	
Static Analysis Of Binary Code With Memory Indirections Using Polyhedra. ⚡	2019/01
C. Ballabriga, J. Forget, L. Gonnord, G. Lipari, J. Ruiz	
In: <i>VMCAI – International Conference on Verification, Model Checking, and Abstract Interpretation</i> , 2019. (regular paper)	
⚡ best paper award	
Working around loops for infeasible path detection in binary programs.	2017/09
J. Ruiz , H. Cassé, M. De Michiel.	
In: <i>SCAM – IEEE International Working Conference on Source Code Analysis and Manipulation</i> , 2017. (regular paper)	
The W-SEPT project: Towards Semantic-aware WCET Estimation.	2017/06
C. Maïza, P. Raymond, C. Parent-Vigouroux, A. Bonenfant, F. Carrier, H. Cassé, P. Cuenot, D. Claraz, N. Halbwachs, F. Carrier, H. Cassé, E. Jahier, H. Li, M. De Michiel, V. Mussot, I. Puaut, C. Rochange, E. Rohou, J. Ruiz , P. Sotin, W.-T. Sun.	
In: <i>WCET – Workshop on Worst-Case Execution Time Analysis</i> , 2017. (regular paper)	
Expressing and Exploiting Path Conflicts in WCET Analysis.	2016/07
V. Mussot, J. Ruiz , P. Sotin, M. De Michiel, H. Cassé.	
In: <i>WCET – Workshop on Worst-Case Execution Time Analysis</i> , 2016. (regular paper)	
Using SMT Solving for the Lookup of Infeasible Paths in Binary Programs.	2015/07
J. Ruiz , H. Cassé.	
In : <i>WCET – Workshop on Worst-Case Execution Time Analysis</i> , 2015. (regular paper)	

Teaching

12h as a teacher assistant in practical work classes at Imperial College London	2019-2020
40h of teaching in practical work classes at the Polytech Lille engineering school	2018-2019
214h (incl. 10h of volunteering) of teaching in practical work classes in university	2014-2017
– <i>Levels taught</i> range from freshman to last year master classes (~15 students each).	
– <i>Topics taught</i> include algorithmics, logic, programming in C, Java, Python, OCaml, Ada, ARM assembly, and compiling techniques and theories.	
– <i>Assisting</i> : conception, supervision and grading of practical exams.	

Languages

French: native.
English: fluent.
Sinographs (Chinese characters): intermediate.

[1] LLVM IR is the intermediate representation of the LLVM compiler toolchain. See: <https://llvm.org>.